

Essential Insights: Week 3.2



सिद्धिमूलं प्रबन्धनम्
भा. प्र. सं. इन्दौर
IIM INDORE

Need for Advanced Prompting

Shift in AI Usage

- Move from simple content generation to decision support
- AI expected to interpret data, evaluate options, and guide action

Challenge with Simple Prompting

- Treats complex problems as single-step tasks
- Produces generic, surface-level recommendations

Business Risk

- Misses trade-offs, root causes, and prioritization
- Inconsistent output quality across attempts

Why Simple Prompts Fall Short

Nature of Business Problems

- Multi-dimensional and interconnected variables
- Require deeper reasoning, not just summarisation

Typical AI Limitations

- Jumps directly to answers without structured reasoning
- Lacks prioritisation and analytical depth

Core Insight

- Effective decision-making requires multi-step thinking

Advanced Prompting: A Structured Approach

From Single-Step to Multi-Step Thinking

- Break problems into stages:
 - Identify patterns
 - Diagnose root causes
 - Evaluate options
 - Recommend actions

Key Benefits

- Improves reasoning quality
- Produces more reliable, explainable outputs

Outcome

- Enables decision-ready insights rather than generic responses

Tree-of-Thoughts (ToT) Prompting

Concept Overview

- Explores multiple reasoning paths simultaneously
- Evaluates and selects the strongest approach

How It Works

- Generate multiple solution branches
- Assess each against defined criteria
- Prune weaker options and refine the best

Key Advantage

- Mimics expert-level strategic thinking

Applications of Tree-of-Thoughts

Strategic Decision-Making

- Evaluate multiple market entry strategies
- Compare risks, feasibility, and alignment

Operational Optimisation

- Explore sourcing or supply chain alternatives
- Balance cost vs risk trade-offs

Financial Analysis

- Simulate different scenarios under uncertainty
- Identify stable and profitable paths

Prompting Techniques Framework

Three Categories of Techniques

- Reasoning Techniques: Chain-of-thought, ToT, decomposition
- Reliability Techniques: Self-consistency, verification, reflection
- Production Techniques: Guardrails, prompt chaining, RAG

Purpose

- Match technique to problem complexity and stakes

Applying the Prompting Framework

Layered Approach

- Start with structured prompt (Role–Task–Context–Constraints–Output)
- Add reasoning techniques for complex problems
- Apply reliability techniques for decision-critical outputs
- Use production techniques for scalability and consistency

Practical Tip

- Start simple, layer only when required

Final Takeaways and Mindset

Advanced Prompting as a Capability

- Enables structured thinking and reliable AI outputs
- Supports high-quality decision-making

Key Differentiator

- Not complexity, but intentional use of techniques

Leadership Mindset

- Know when to use which technique
- Continuously refine through practice

Next Step

- Apply advanced prompting to real business problems